

Test Report No:	XXXXXXXXXX	Page 1 of 30
		Issue date XXXXX
Manufacturer:		
Test item:	Standalone LED Modules for General Lighting	
Identification:	Serial No.:	
Receipt No.:	Date of receipt:	
Testing laboratory and its address:		
Test specification:	IS 16103 (Part 1):2012 +A1:2014	
Test Result:	The test item passed/failed the test specification(s).	
Other Aspects: This report consists of xx pages and an attachment:		

Tested by:	Approved by / Authorized Signatory:	Issued by:
(Name / Designation)	(Name / Designation)	(Name / Designation)
Date:	Date:	Date:

TEST REPORT
IS 16103 (Part 1)
LED modules for general lighting
Part 1: Safety requirements

Report Number. :

Date of issue :

Total number of pages

Applicant's name :

Address :

Test specification:

Standard : IS 16103 (Part 1):2012 +A1:2014

Test procedure..... : Compliance Report

Non-standard test method..... : N/A

Test Report Form No..... : BIS_LUM/EL_16103-1_V1.0

Test Report Form(s) Originator.... : Bureau of Indian Standards

Master TRF : 20/04/2020

Test item description.....:

Trade Mark.....:

Manufacturer.....:

Model/Type Reference.....:

Rating.....:

Tested by:	Approved by / Authorized Signatory:	Issued by:
(Name / Designation)	(Name / Designation)	(Name / Designation)
Date:	Date:	Date:

Report No.:

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Dated:

Description	Measurement/ testing	Total No. of tests	Total no. of applicable tests/ Req.	No. of tests/ Req. passed	Page No.
Marking requirements	Marking				
Mechanical properties	Terminals				
Insulating properties	Provision for protective earthing				
Insulating properties	Protection against accidental contact with live parts				
Insulating properties	Moisture resistance and insulation				
Insulating properties	Electric strength				
Insulating properties	Fault conditions				
Mechanical Properties	Construction				
Safety requirements	Creepage distances and clearances				
Mechanical Properties	Screws, current –carrying parts and connections				
Safety requirements	Resistance to heat, fire and tracking				
Safety requirements	Resistance to corrosion				

Certificate: It is certified that the above tests were performed and found to be passing in the requirement test

.....
(Approving Authority)

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Dated:

Copy of marking plate:

Copy of trademark:

(Provided on the equipment)

Table – List of Attachments

Attachment No.	Attachment Description	No. of pages in Attachment
Attachment – 1	Photo Document	Page no.

General remarks:

The test results presented in this report relate only to the object tested.

This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory.

Possible test case verdicts:

- test case does not apply to the test object.....: N/A
- test object does meet the requirement.....: P (Pass)
- test object does not meet the requirement.....: F (Fail)

Testing.....:

Date of receipt of test item.....:

Date (s) of performance of tests.....:

Laboratory conditions.....:

Ambient Temperature.....:

Ambient Humidity.....:

Dated:

General Product Information:-**Product Description:****Model Tested :**

Sr. No.	Product description	Model	Specifications
1			

Representative Models:

Sr. No.	Product description	Model	Specifications	Variation From Family Representative

Supply connections:**Condition of sample at the time of receipt :****Technical Considerations:****Report Summary:**

Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
4	GENERAL REQUIREMENTS			
4.4	Integral modules treated as part of luminaires defined in clause 0.5 of IS 10322 (Part 1)	--	Yes ☐ No ☐	—
4.5	Independent modules complies with requirements of IS 10322 (Part 1)	--	Yes ☐ No ☐	—

Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
5	GENERAL TEST REQUIREMENTS			
5.5	SELV-operated LED modules comply with Annex L of IS 15885 (Part 1)	--	Yes ☐ No ☐	—

Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
6	CLASSIFICATION OF LUMINAIRES			
a)	Built-in module	--	Yes ☐ No ☐	—
b)	Independent module	--	Yes ☐ No ☐	
c)	Integral module	--	Yes ☐ No ☐	—
	For Integral module; Note to 1.2.1 of IS 10322 (Part 1) applies..	--		—

Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
	MARKING			
7.1	Mandatory markings:	--		
a)	mark of origin*	--		
b)	model number, type reference*	--		
c1)	Power or power range			
c2)	constant voltage module; rated supply voltage and supply frequency	--		
c3)	constant current module; rated supply current and supply frequency	--		
d)	nominal power*	--		
e)	indication of connections, wiring diagram*	--		
f)	value of t_c *	--		
g)	eye protection*	--		
h)	marking of built-in modules only*	--		
j)	heat transfer temperature t_d *	--		
k)	power for heat-conduction P_d *	--		
7.2	Location of marking*	--		
	- marking of 7.1 (a) to 7.1 (c) and 7.1 (f) shall be marked on the module.*	--		
	- marking of 7.1 (d) to 7.1 (h) and 7.1 (k) shall be marked legible on the module or on the module data sheet*	--		
	- integral modules a) to g) in literature*	--		
7.3	Durable and legibility of marking	--		
	- marking of a), b), c) and f) legible after test rubbing 15 s water	--		
	- marking of d) to h) and k) complies is checked by inspection*	--		

*Total number of Requirements to be observed / inspected =

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Total number of tests to be conducted =

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
8	Terminals			
	Screw terminals according Section 14 of IS 10322 (Part 1):			
	Separately approved; component list	--	(see Annex 2)	
	Part of the luminaire	--	(see Annex 3)	
	Screw less terminals according section 15 of IS 10322 (Part 1):			
	Separately approved; component list	--	(see Annex 2)	
	Part of the luminaire	--	(see Annex 4)	

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
9	PROVISIONS FOR PROTECTIVE EARTHING			
	Provisions for protective earthing*	--		
	Terminal complying with clause 8	--		
	Locked against loosening and not possible to loosen by hand	--		
	Not possible to loosen clamping means unintentionally on screwless terminals	--		
	Earthing via means of fixing*	--		
	Earthing terminal only used for the earthing of the control gear*	--		
	All parts of material minimizing the danger of electrolytic corrosion	--		
	Made of brass or equivalent material*	--		
	Contact surface bare metal*	--		
	Provision for functional earthing*	--		
	Comply with clause 8	--		
	Earth contact via the track on the printed board	--		
	Test with a current of 25 A between earthing terminal and each of the accessible metal parts; measured resistance (Ω) at 10 A according 7.2.3 of IS 10322-1: < 0,5 Ω	--		
	Earthing of built-in lamp controlgear*	--		
	Earth by means of fixing to earthed metal of luminaire in compliance of 7.2 of IS 10322-1	--		
	Earthing terminal only for earthing the built-in controlgear*	--		
	Earthing via independent controlgear*	--		
	Earth connection to other equipment*	--		
	Looping or through connection, conductor min. 1,5mm ² and of copper or equivalent	--		
	Protective earthing wires in line with 5.3.1.1 and clause 7*	--		
	Earthing of the lamp compartments powered via the independent lamp controlgear*	--		

	Test with a current of 25 A between input and output earth terminals; measured resistance (Ω) between earthing terminal and each of the accessible metal parts at 10 A according 7.2.3 of IS 10322-1: < 0,5 Ω	--		
--	---	----	--	--

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
10	PROTECTION AGAINST ACCIDENTAL CONTACT WITH LIVE PARTS			
- (10.1)	Controlgear protected against accidental contact with live parts	--		
- (10.1)	Lacquer or enamel not used for protection or insulation*	--		
	Adequate mechanical strength on parts providing protection	--		
- (10.2)	Capacitors > 0,5 μ F: voltage after 1 min (V): < 50 V	--		
- (10.3)	Controlgear providing SELV	--		
	Accessible conductive parts are insulated from live parts by double or reinforced insulation in SELV controlgear*	--		
	No connection between output circuit and the body or protective earthing circuit	--		
	No possibility of connection between output circuit and the body or protective earthing circuit through other conductive parts	--		
	SELV outputs separated by at least basic insulation*	--		
	ELV conductive parts insulated as live parts*	--		
	Tests according Annex M of IS 15885-1*	--		
- (10.4)	Accessible conductive parts in SELV circuits	--		
	Output voltage under load ≤ 25 V r.m.s. or ≤ 60 V d.c.	--		
	If output voltage > 25 V r.m.s. or > 60 V d.c.; No load output ≤ 35 V peak or ≤ 60 V d.c and touch current does not exceed 0,7 mA (peak) or 2 mA d.c.	--		
	One conductive part is insulated if output voltage or current exceeding the values above and withstand test voltage 500 V	--		
	Double or reinforced insulation bridged by appropriate and at least two resistors or two Y2 capacitors or one Y1 capacitor	--		

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	Y1 or Y2 capacitors comply with ISQC 302400*	--		
	Resistors comply with test (a) in 14.1 of IS 616*	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
11	MOISTURE RESISTANCE AND INSULATION			
	After storage 48 h at 91-95% relative humidity and 20-30 °C measuring of insulation resistance with d.c. 500 V (MΩ):	--		
	For basic insulation $\geq 2 \text{ M}\Omega$	--		
	For double or reinforced insulation $\geq 4 \text{ M}\Omega$	--		
	Between primary and secondary circuits in controlgear providing SELV, values in Annex L in IEC 61347-1	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
12	Electric Strength			
	Immediately after clause 11 electric strength test for 1 min	--		
	Working voltage ≤ 42 V, test voltage 500 V	--		
	Working voltage > 42 V ≤ 1000 V, test voltage (V):	--		
	Basic insulation, $2U + 1000$ V	--		
	Supplementary insulation, $2U + 1000$ V	--		
	Double or reinforced insulation, $4U + 2000$ V	--		
	No flashover or breakdown*	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
13	FAULT CONDITIONS			
- (14)	When operated under fault conditions the controlgear:	--		
	- does not emit flames or molten material*	--		
	- does not produce flammable gases*	--		
	- protection against accidental contact not impaired	--		
	Thermally protected controlgear does not exceed the marked temperature value	--		
	Fault conditions: capacitors, resistors or inductors without proof of compliance with relevant specifications have been short-circuited or disconnected	--		
- (14.1)	Short-circuit of creepage distances and clearances if less than specified in clause 16 in Part 1 (except between live parts and accessible metal parts)	--		
	Creepage distances on printed boards less than specified in clause 16 in Part 1 provided with coating according to IS 15382 (Part 3)	--		
- (14.2)	Short-circuit or interruption of semiconductor devices	--		
- (14.3)	Short-circuit across insulation consisting of lacquer, enamel or textile	--		
- (14.4)	Short-circuit across electrolytic capacitors	--		
- (14.5)	After the tests has been carried out on three samples:	--		
	The insulation resistance $\geq 1 \text{ M}\Omega$	--		
	No flammable gases*	--		
	No accessible parts have become live	--		
	During the tests, a five-layer tissue paper, where the test specimen is wrapped, does not ignite*	--		
	Temperature declared thermally protected lamp controlgear fulfil requirements in Annex C	--		

13.2	Overpower condition	--		
	Module withstands overpower condition >15 min.	--		
	Module with automatic protective device or power limiter, test performed 15 min. at limit.	--		
	No fire, smoke or flammable gas is produced*	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
15	Construction			
	Wood, cotton, silk, paper and similar fibrous material not used as insulation*	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
16	CREEPAGE DISTANCES AND CLEARANCES			
	Creepage and distances and clearances in compliance with section 11 of IS 10322 (Part 1)	--		
	Class of protection	--		
	Working voltage (V)	--		
	Voltage form	--	Sinusoidal ☐ / Non-sinusoidal ☐	
	PTI	--	< 600 ☐ / 600 ☐	
	Impulse withstand category(Normal category II) (Category III Annex U)	--	Category II ☐ Category III ☐	
	Rated pulse voltage (kV)	--		
	(1) Live parts of different polarity: cr (mm); cl (mm)	--		
	(2) Live parts and accessible parts: cr (mm); cl (mm)	--		
	(3) Parts becoming live: cr (mm); cl (mm)	--		
	(4) Outer surface of cable: cr (mm); cl (mm)	--		
	(5) Live parts of switches: cr (mm); cl (mm)	--		
	(6) Live parts and supporting surface: cr (mm); cl (mm)	--		

Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
17	SCREWS, CURRENT-CARRYING PARTS AND CONNECTIONS	--		
	Cl. 17 refer to Cl. 17 of IS 15885 (Part 1) which refer to Cl. 4.11 and 4.12 of IS 10322 (Part 1)(clause numbers between parentheses refer to IS 10322 (Part 1)	--		
(4.11)	Electrical connections*	--		
(4.11.1)	Contact pressure*	--		
(4.11.2)	Screws:*	--		
	- self-tapping screws*	--		
	- thread-cutting screws*	--		
(4.11.3)	Screw locking:*	--		
	- spring washer*	--		
	- rivets*			
(4.11.4)	Material of current-carrying parts*	--		
(4.11.5)	No contact to wood or mounting surface*	--		
(4.11.6)	Electro-mechanical contact systems	--		
(4.12)	Mechanical connections and glands	--		
(4.12.1)	Screws not made of soft metal*	--		
	Screws of insulating material*	--		
	Torque test: torque (Nm); part	--		
	Torque test: torque (Nm); part	--		
	Torque test: torque (Nm); part	--		
(4.12.2)	Screws with diameter < 3 mm screwed into metal	--		
(4.12.4)	Locked connections:	--		
	- fixed arms; torque (Nm)	--		
	- lampholder; torque (Nm)	--		

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Dated:

	- push-button switches; torque 0,8 Nm	--		
(4.12.5)	Screwed glands; force (Nm)	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
18	RESISTANCE TO HEAT, FIRE AND TRACKING			
(18.1)	Ball-pressure test:	--		
	-part tested; temperature (°C)	--		
(18.2)	Test of printed boards:	--		
	-part tested :	--		
	-part tested :	--		
(18.3)	Glow-wire test (650°C):	--		
	-part tested :	--		
	-part tested :	--		
(18.4)	Needle flame test (10 s):	--		
	-part tested :	--		
	-part tested :	--		
(18.5)	Tracking test:	--		
	-part tested :	--		
	-part tested :	--		

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Clause No	Test / Requirement name	Code	Test result/ observation	Verdict
19	RESISTANCE TO CORROSION			
	- test according 4.18.1 of IS 10322 (Part 1)	--		
	- adequate varnish on the outer surface*	--		

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A	ANNEX A - TESTS	
	All tests performed in accordance with the advice given in Annex H of IS 15885 (Part 1), if applicable	

13 (14)	TABLE: tests of fault conditions	
Part	Simulated fault	Hazard
		YES/NO
		YES/NO
		YES/NO
		YES/NO
		YES/NO
		YES/NO

18 (18.1)	TABLE: Ball Pressure Test of Thermoplastics			
Allowed impression diameter (mm) :				—
Object/ Part No./ Material	Manufacturer/ trademark	Test temperature (°C)	Impression diameter (mm)	
Supplementary information:				

18 (18.3)	TABLE: Glow-wire test					
Glow wire temperature :				650°C		—
Object/ Part No./ Material	Manufacturer/ trademark	Duration of application of test flame (ta); (s)	Ignition of specified layer Yes/No	Duration of burning (tb) (s)	Verdict	

Dated:

Any flame or glowing of the sample extinguished within 30 s of withdrawing the glow-wire, and any burning or molten drop did not ignite the underlying parts (Yes/No)					
Supplementary information:					

18 (18.4)	TABLE: Needle-flame test					
Object/ Part No./ Material	Manufacturer/ trademark	Duration of application of test flame (ta); (s)	Ignition of specified layer Yes/No	Duration of burning (tb) (s)	Verdict	
Supplementary information:						

18 (18.5)	TABLE: Proof tracking test					
Test voltage PTI		175 V			—	
Object/ Part No./ Material	Manufacturer/ trademark	Withstand 50 drops without failure on three places or on three specimens			Verdict	
Supplementary information:						

ANNEX 1	SELV-operated LED modules			
	Cl. 5.5 refer to ANNEX I of IS 15885 Part 2 sec 13 which refer to ANNEX L of IS 15885 (Part 1) (clause numbers between parentheses refer to ANNEX L of IS 15885 (Part 1))			—
(M.3)	Classification			
	Class I	Yes ☐	No ☐	—
	Class II	Yes ☐	No ☐	—
	Class III	Yes ☐	No ☐	—
	non-inherently short circuit proof controlgear	Yes ☐	No ☐	—
	inherently short circuit proof controlgear	Yes ☐	No ☐	—
	fail safe controlgear	Yes ☐	No ☐	—
	non-short-circuit proof controlgear	Yes ☐	No ☐	—

(M.4)	Marking		
	Adequate symbols are used		
(M.5)	Protection against electric shock		
	Comply with 9.2 of IEC 61558-1		
(M.6)	Heating		
	No excessive temperatures in normal use		
	Value if capacitor tc marked		—
	Winding insulation classified as Class		—
	Comply with tests of clause 14 of IEC 61558-1 with adjustments		
(M.7)	Short-circuit and overload protection		
	Comply with tests of clause 15 of IEC 61558-1 with adjustments		
(M.8)	Insulation resistance and electric strength		
(M.8.1)	Conditioned 48 h between 91 % and 95 %		
(M.8.2)	Insulation resistance		
	Between input- and output circuits not less than 5 MΩ		
	Between metal parts of class II convertors which are separated from live parts by basic insulation only and the body not less than 5 MΩ		
	Between metal foil in contact with the inner and outer surfaces of enclosures of insulating material not less than 2 MΩ		
(M.8.3)	Electric strength		
	1) Between live parts of input circuits and live parts of output circuits		
	2) Over basic or supplementary insulation between:		
	a) live parts having different polarity		
	b) live parts and body if intended to be connected to protective earth		
	c) accessible metal parts and a metal rod of the same diameter as the flexible cable or cord		
	d) live parts and an intermediate metal part		
	e) intermediate metal parts and the body		
	f) each input circuit and all other input circuits		
	3) Over reinforced insulation between the body and live parts		

(M.9)	Construction		
(M.9.1)	Transformer comply with 19.12 of IEC 61558-1 and 19 of IEC 61558-2-6		
	HF transformer comply with 19 of IEC 61558-2-16		
(M.10)	Components		
	Protective devices comply with 20.6 – 20.11 of IEC 61558-1		
(M.11)	Creepage distances, clearances and distances through insulation		
	Creepage distances and clearances not less than in Clause 16		
	Distance through insulation according Table L.5 in IS 15885-1		
	1) Basic distance through insulation		
	Required distance (mm)		—
	Measured (mm)		
	Supplementary information		—
	2) Supplementary distance through insulation		
	Required distance (mm)		—
	Measured (mm)		
	Supplementary information		—
	3) Reinforced distance through insulation		
	Required distance (mm)		—
	Measured (mm)		
	Supplementary information		—

ANNEX 2	TABLE: Critical components information					
Object / part No.	Code	Manufacturer/ trademark	Type/model	Technical data	Standard	Mark(s) of conformity

Supplementary information: N/A

¹⁾ Provided evidence ensures the agreed level of compliance. See OD-CB2039.

The codes above have the following meaning:

- A - The component is replaceable with another one, also certified, with equivalent characteristics
- B - The component is replaceable if authorised by the test house
- C - Integrated component tested together with the appliance
- D -Alternative component

ANNEX 3	Screw terminals (part of the luminaire)		
(14)	SCREW TERMINALS		
(14.2)	Type of terminal..... :		—
	Rated current (A)..... :		—
(14.3.2.1)	One or more conductors		
(14.3.2.2)	Special preparation		
(14.3.2.3)	Terminal size		
	Cross-sectional area (mm ²)..... :		—
(14.3.3)	Conductor space (mm)..... :		
(14.4)	Mechanical tests		
(14.4.1)	Minimum distance		
(14.4.2)	Cannot slip out		
(14.4.3)	Special preparation		
(14.4.4)	Nominal diameter of thread (metric ISO thread) :	M	
	External wiring		
	No soft metal		
(14.4.5)	Corrosion		
(14.4.6)	Nominal diameter of thread (mm) :		
	Torque (Nm) :		
(14.4.7)	Between metal surfaces		
	Lug terminal		
	Mantle terminal		
	Pull test; pull (N) :		
(14.4.8)	Without undue damage		

ANNEX 4	Screwless terminals (part of the luminaire)		
(15)	SCREWLESS TERMINALS		
(15.2)	Type of terminal..... :		—
	Rated current (A)..... :		—
(15.3.1)	Material		
(15.3.2)	Clamping		
(15.3.3)	Stop		
(15.3.4)	Unprepared conductors		
(15.3.5)	Pressure on insulating material		
(15.3.6)	Clear connection method		
(15.3.7)	Clamping independently		
(15.3.8)	Fixed in position		
(15.3.10)	Conductor size		
	Type of conductor		
(15.5.1)	Terminals internal wiring		
(15.5.1.1)	Pull test spring-type terminals (4 N, 4 samples) :		
(15.5.1.2)	Pull test pin or tab terminals (4 N, 4 samples) :		
	Insertion force not exceeding 50 N		
(15.5.1.2)	Permanent connections: pull-off test (20 N)		
(15.5.2)	Electrical tests		
	Voltage drop (mV) after 1 h (4 samples) :		
	Voltage drop of two inseparable joints		
	Number of cycles:		—
	Voltage drop (mV) after 10th alt. 25th cycle (4 samples)..... :		
	Voltage drop (mV) after 50th alt. 100th cycle (4 samples)..... :		
	After ageing, voltage drop (mV) after 10th alt. 25th cycle (4 samples) :		
	After ageing, voltage drop (mV) after 50th alt. 100th cycle (4 samples) :		
(15.6)	Terminals and connections for external wiring		
(15.6.1)	Conductors		
	Terminal size and rating		
(15.6.2)	Mechanical tests		

Dated:

(15.6.2.1)	Pull test spring-type terminals or welded connections (4 samples); pull (N)		
(15.6.2.2)	Pull test pin or tab terminals (4 samples); pull (N)		
(15.6.3)	Electrical tests		
	Tests according 15.6.3.1 + 15.6.3.2 in IEC 60598-1		

(15.6.3.1) (15.6.3.2)	TABLE: Contact resistance test / Heating tests										
	Voltage drop (mV) after 1 h										—
terminal	1	2	3	4	5	6	7	8	9	10	
voltage drop (mV)											
	Voltage drop of two inseparable joints										
	Voltage drop after 10th alt. 25th cycle										
	Max. allowed voltage drop (mV) :										—
terminal	1	2	3	4	5	6	7	8	9	10	
voltage drop (mV)											
	Voltage drop after 50th alt. 100th cycle										
	Max. allowed voltage drop (mV) :										—
terminal	1	2	3	4	5	6	7	8	9	10	
voltage drop (mV)											
	Continued ageing: voltage drop after 10th alt. 25th cycle										
	Max. allowed voltage drop (mV) :										—
terminal	1	2	3	4	5	6	7	8	9	10	
voltage drop (mV)											
	Continued ageing: voltage drop after 50th alt. 100th cycle										
	Max. allowed voltage drop (mV) :										—
terminal	1	2	3	4	5	6	7	8	9	10	
voltage drop (mV)											
Supplementary information:											

*** END OF TEST REPORT ***